

Chapter 17

Unemployment: Search and Efficiency Wages

Chapter 17 Topics

- Behavior of the unemployment rate, participation rate, employment/population ratio.
- Search model of unemployment.
- Efficiency wage model.

Key Determinants of the Unemployment Rate

- Aggregate economic activity
- Demographics
- Government intervention
- Sectoral shifts

Figure 17.1

The U.S. Unemployment Rate, 1948–2009

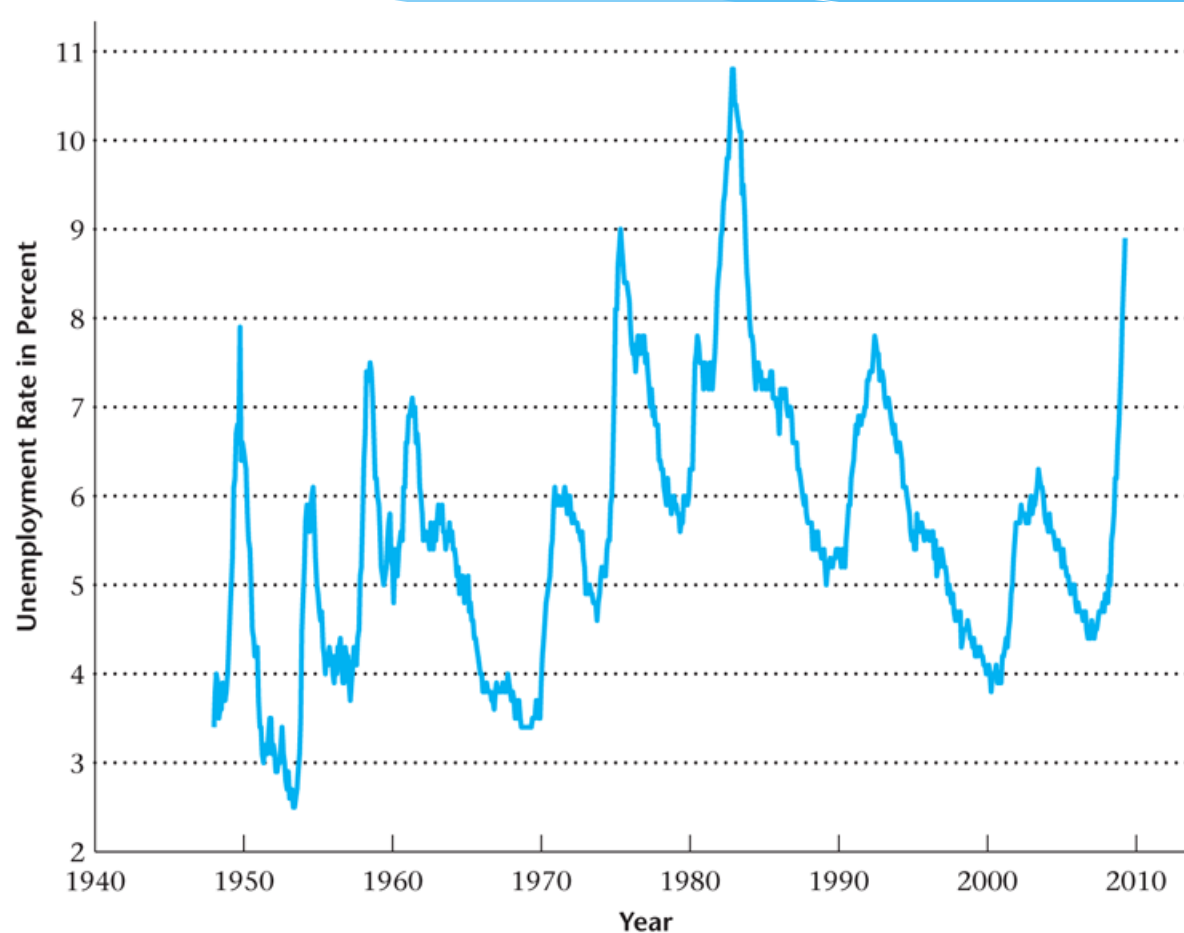


Figure 17.2

Deviations from Trend in the Unemployment Rate
and Percentage Deviations from Trend in Real GDP

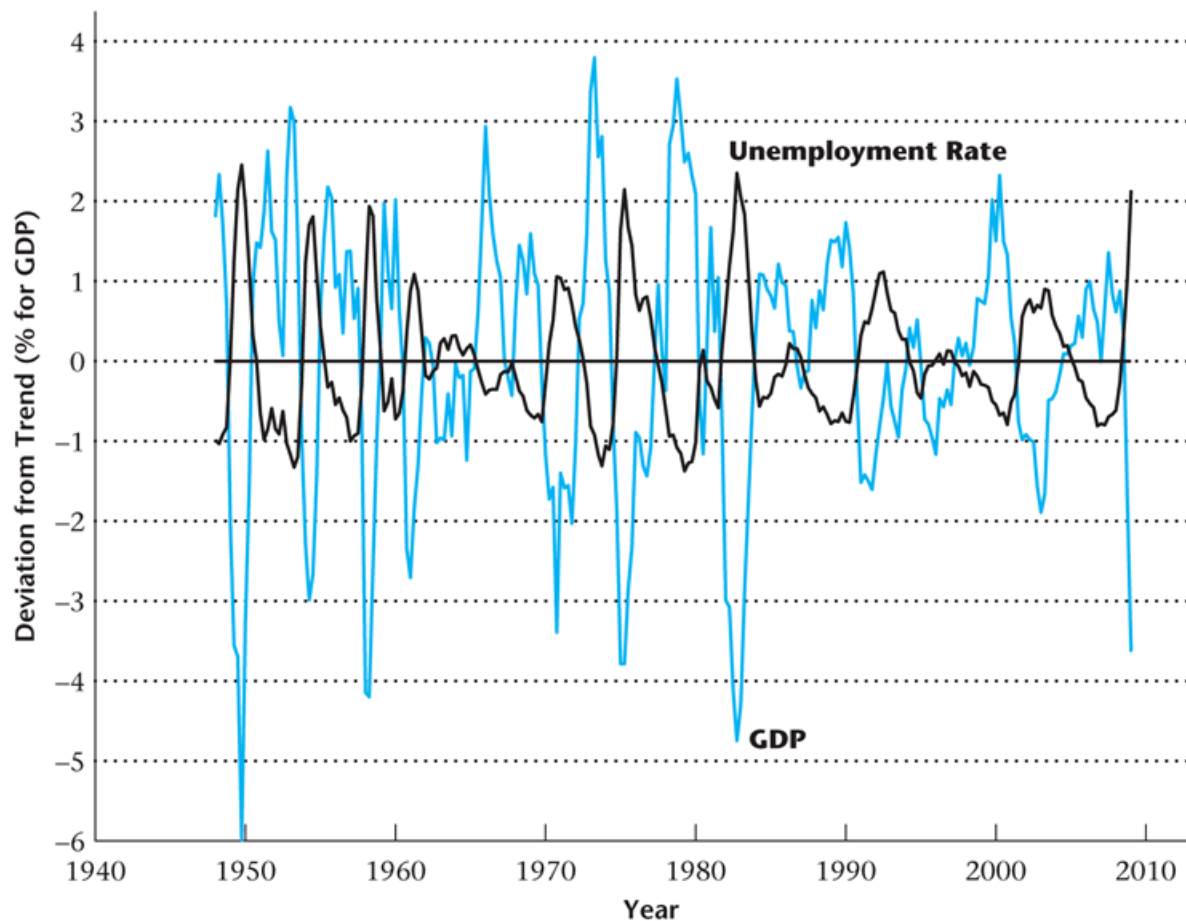


Figure 17.3

The U.S. Participation Rate

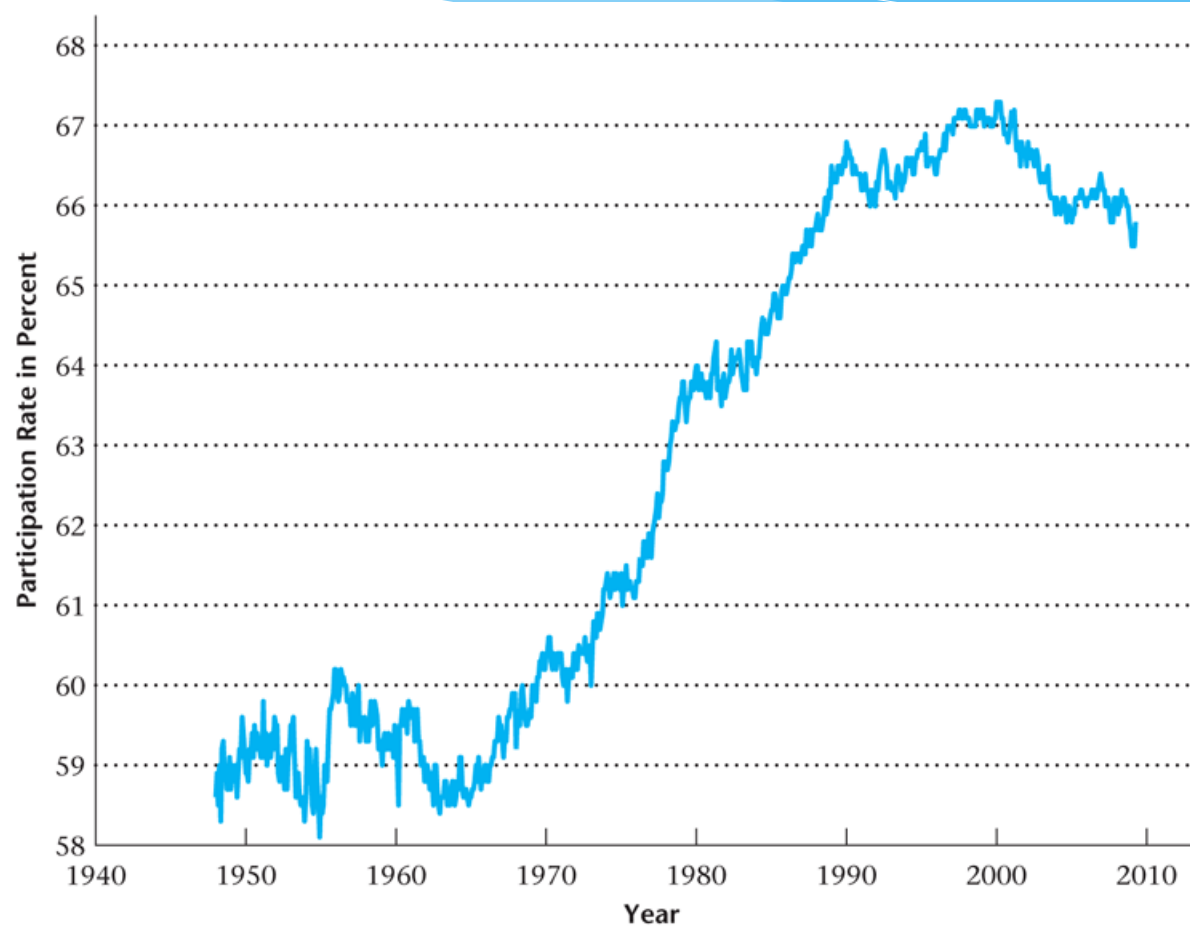


Figure 17.4

Labor Force Participation of Men and Women

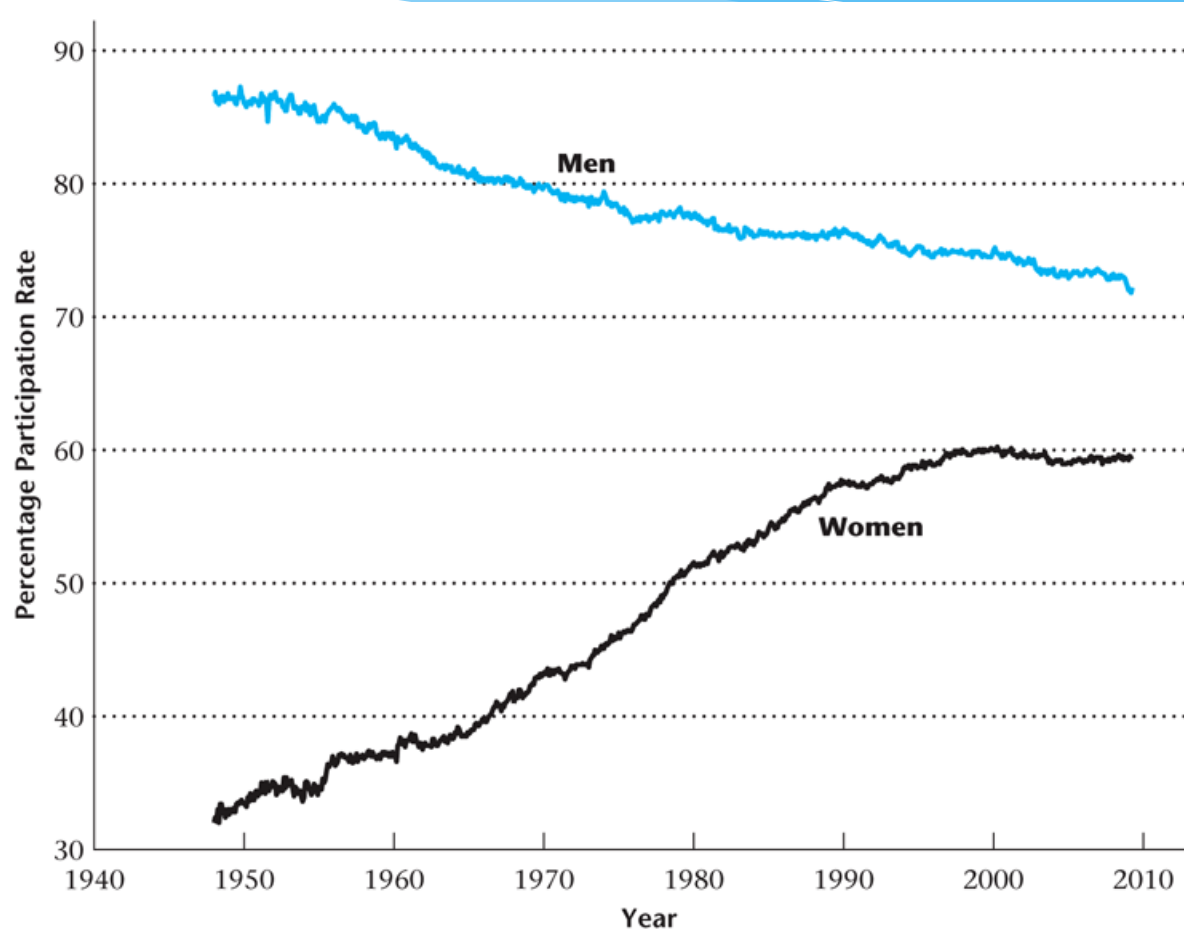
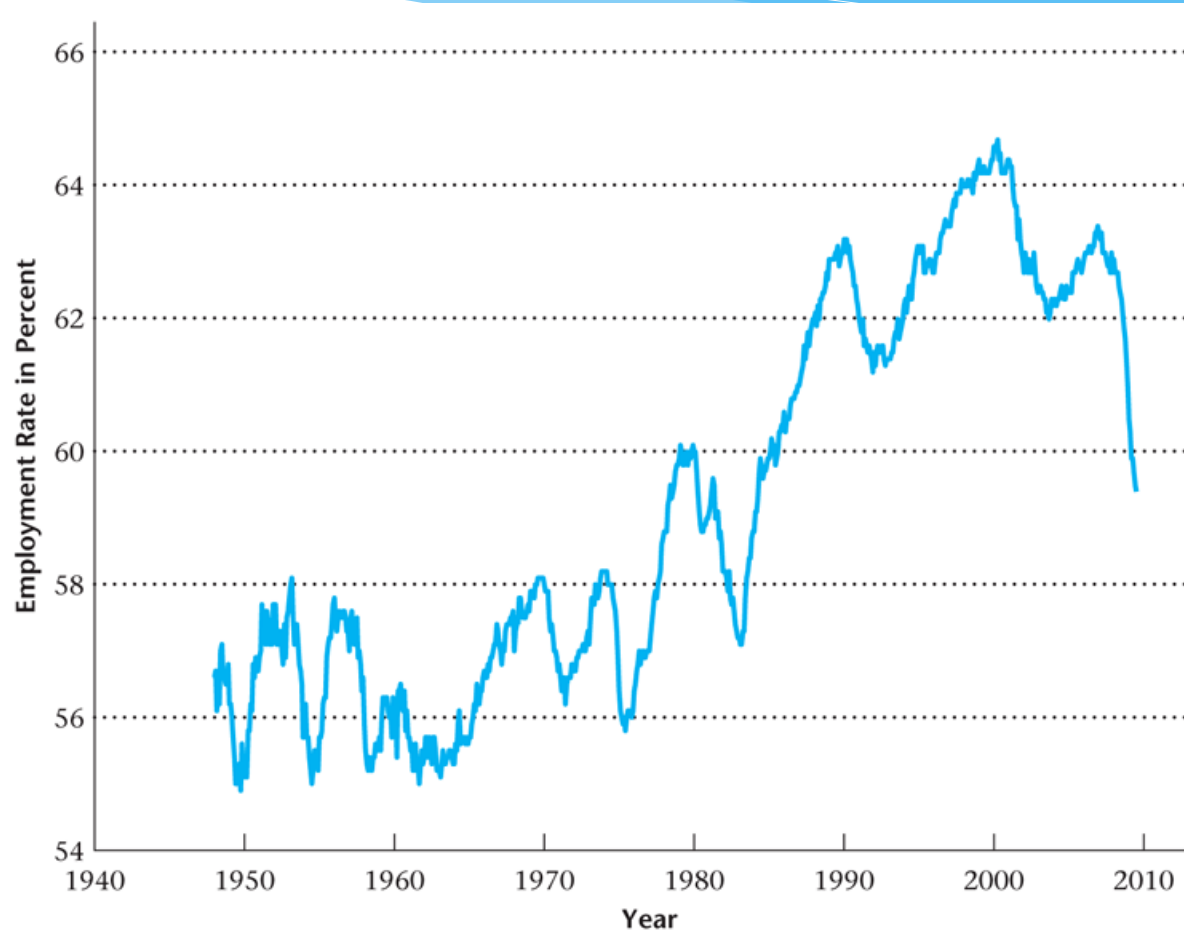


Figure 17.5

The Employment/Population Ratio for the United States



Search Model of Unemployment

- Unemployed workers receive job offers to work at a particular wage.
- When an unemployed worker receives a job offer, he she has to decide whether to accept it or to reject it and continue searching for work.
- Assume that workers are separated from their jobs at a constant rate.
- Want to analyze the determinants of the *reservation wage* – the lowest wage offer that an unemployed worker is willing to accept.

Figure 17.6

The Welfare of an Employed Worker

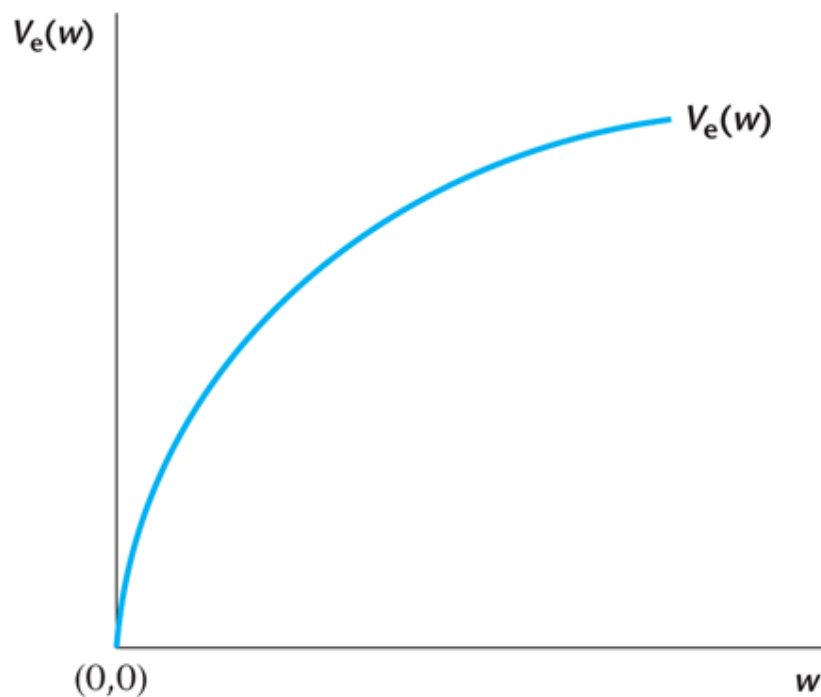
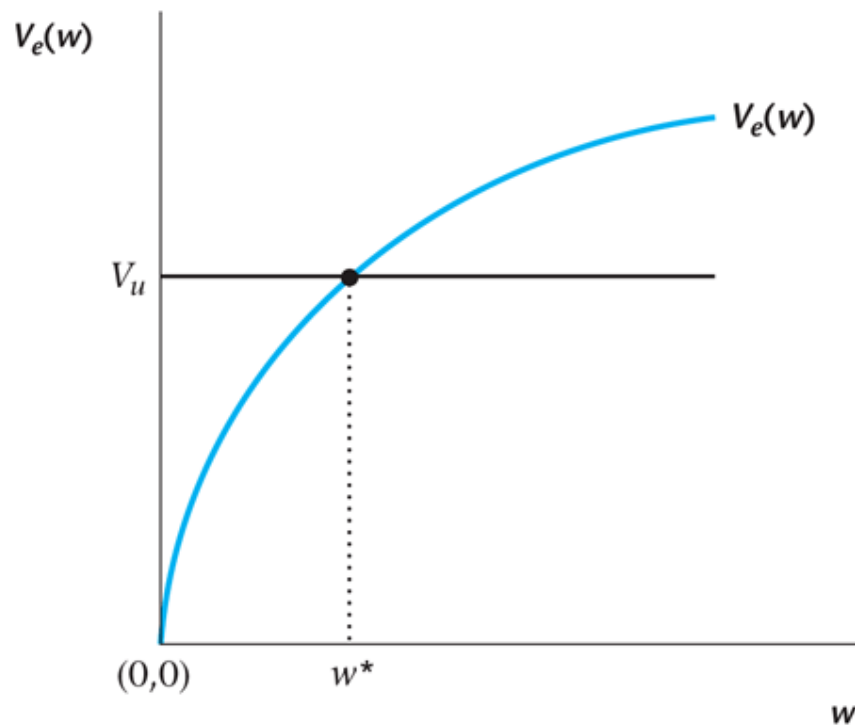


Figure 17.7

The Reservation Wage



The Reservation Wage

- Increases when the unemployment insurance benefit b increases.
- Increases if wage income is taxed at a higher rate.

Figure 17.8

An Increase in the Unemployment Insurance Benefit b

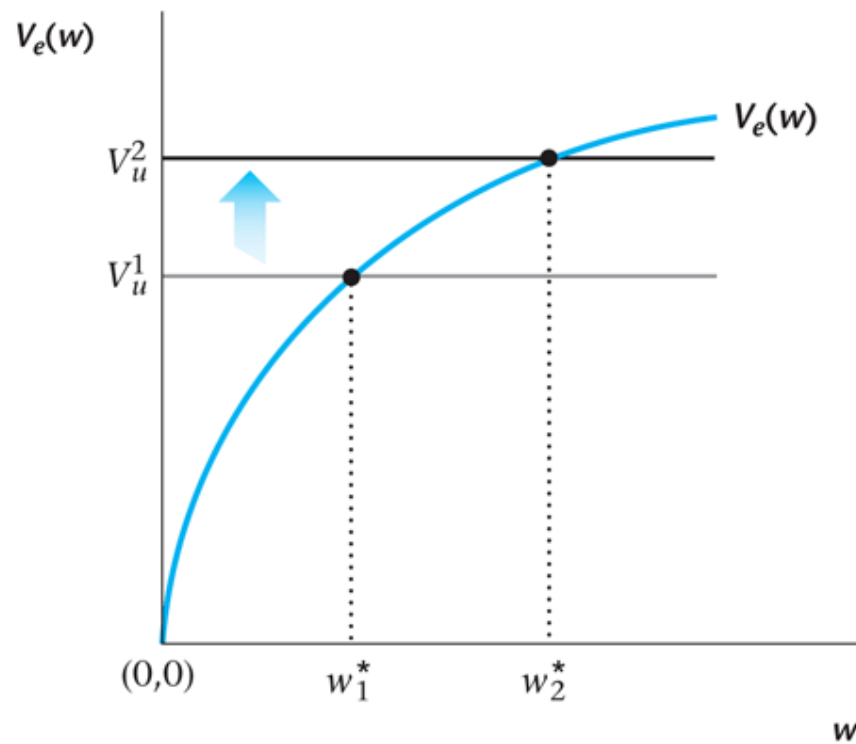
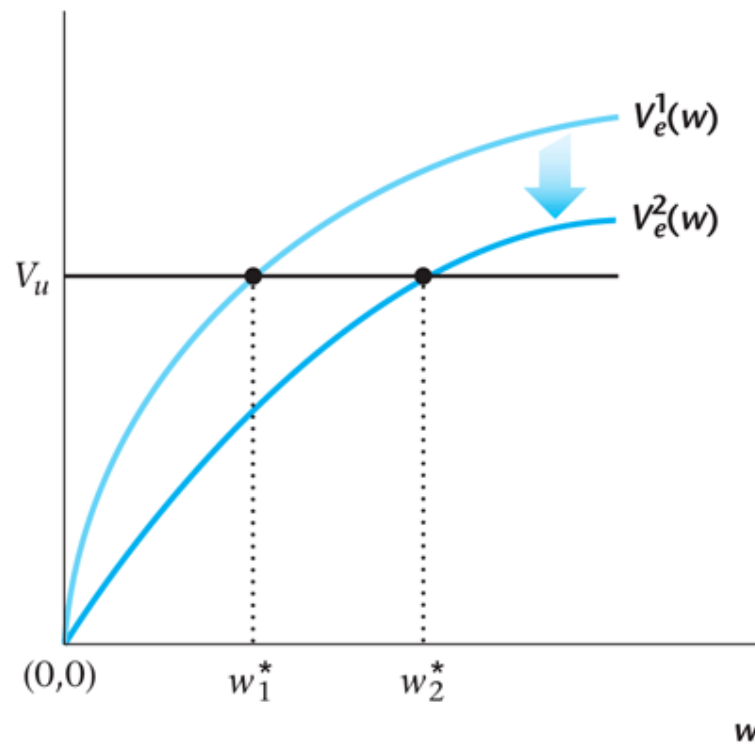


Figure 17.9

An Increase in the Taxes on the Wage Income of the Employed



Determining the Long-Run Unemployment Rate

The unemployment rate increases when the reservation wage increases, the separation rate increases, or the rate at which unemployed workers receive job offers falls.

Long-Run Unemployment Rate

Equation determining the long-run unemployment rate:

$$s(1 - U) = UpH(w^*)$$

Figure 17.10

The Fraction of Unemployed Workers Receiving a Wage Offer Greater than w

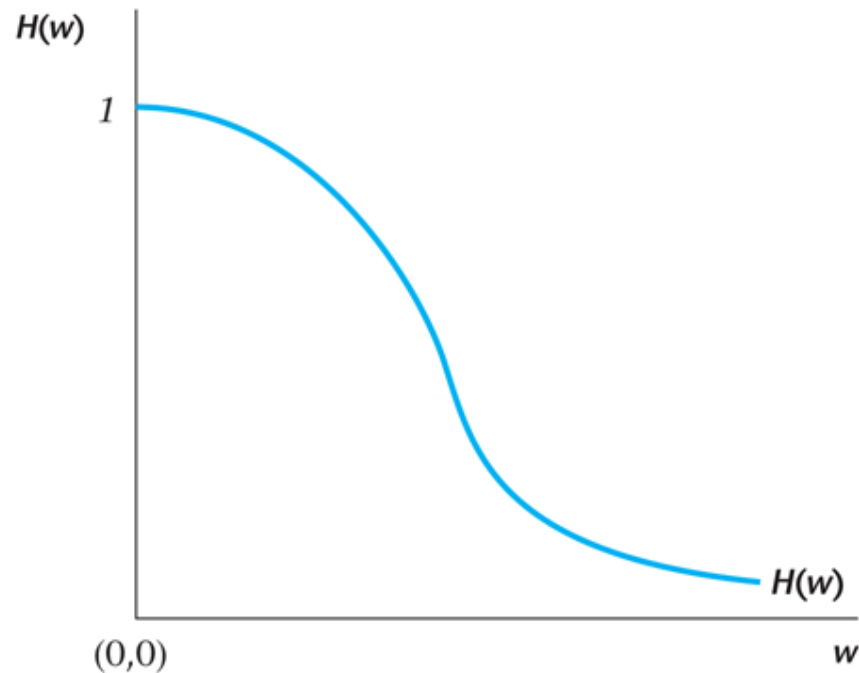


Figure 17.11

The Determination of the Unemployment Rate U^* in the Search Model

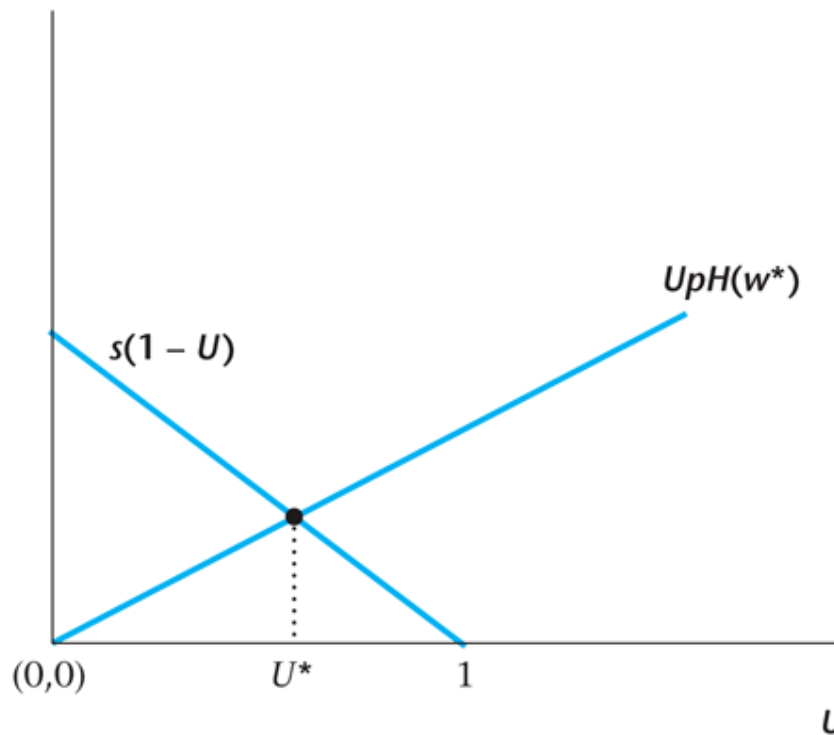


Figure 17.12

The Determination of the Reservation Wage and the Unemployment Rate in the Search Model

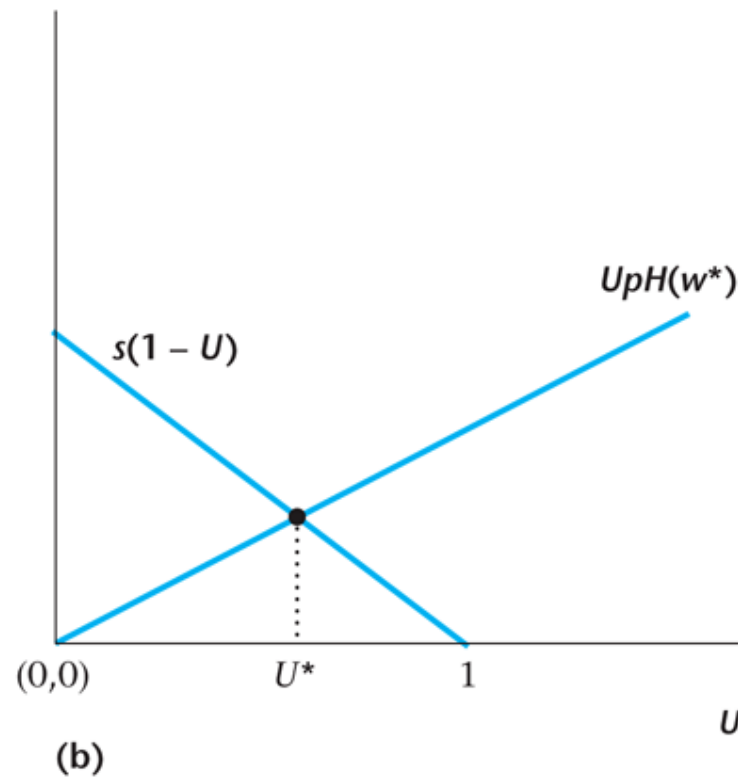
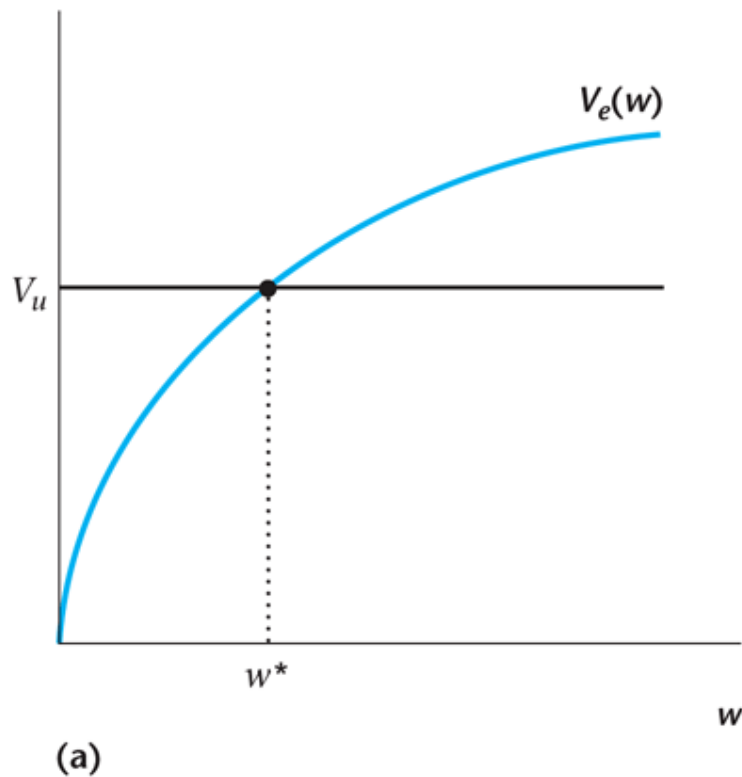
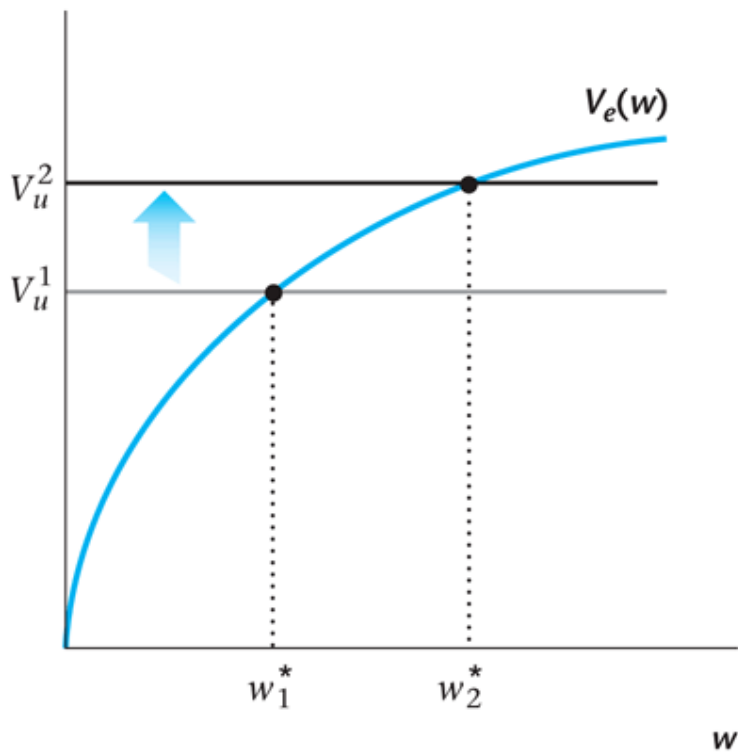
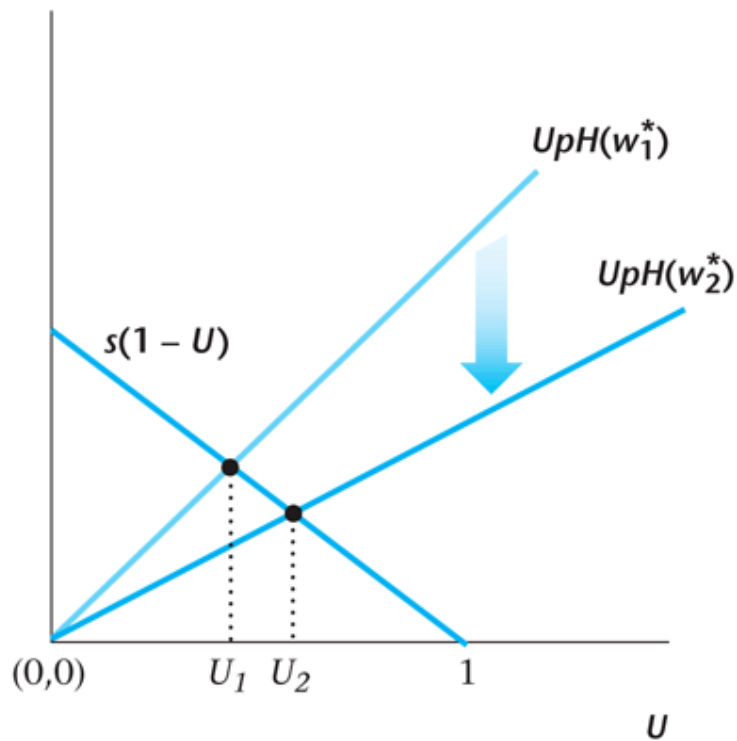


Figure 17.13

An Increase in the Unemployment Insurance Benefit b



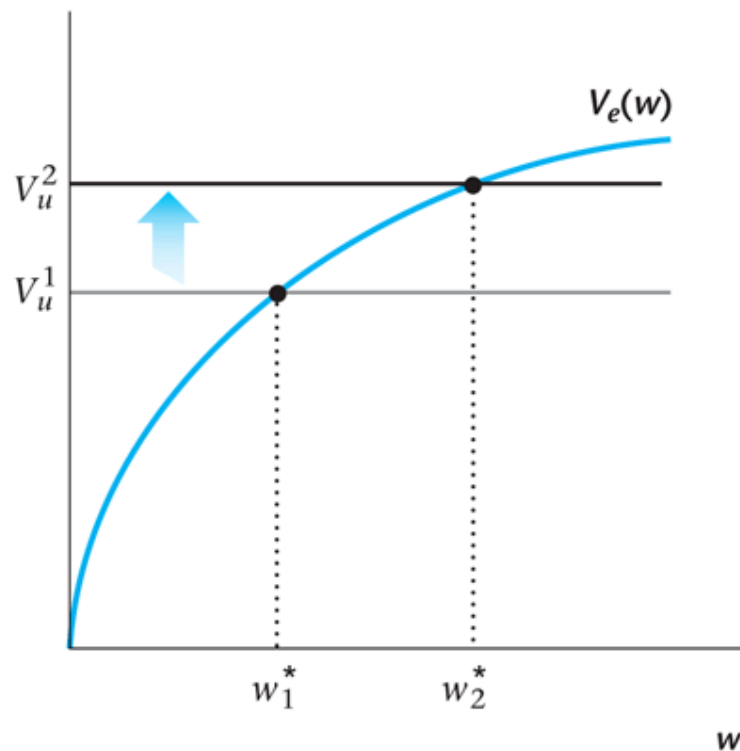
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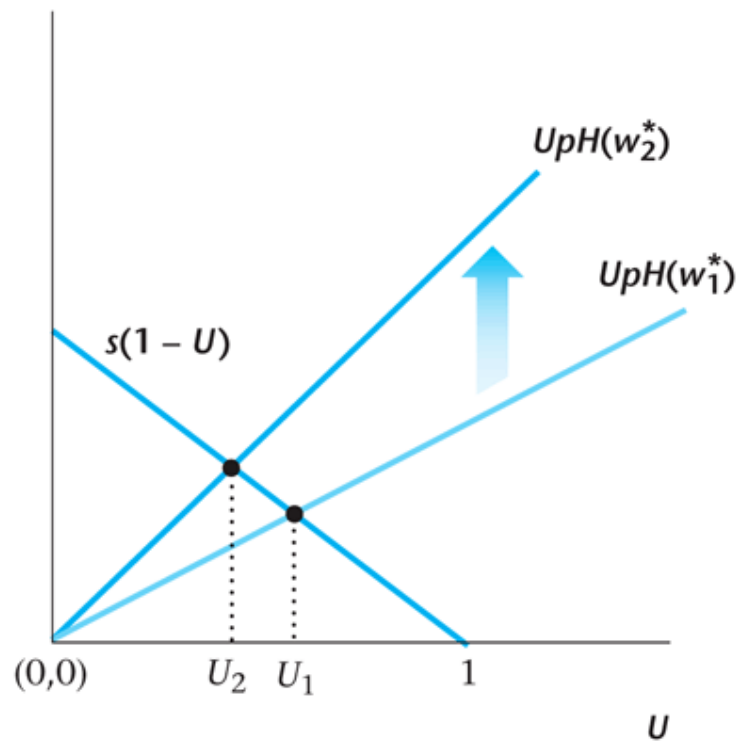
(b)

Figure 17.14

An Increase in the Job Offer Rate p



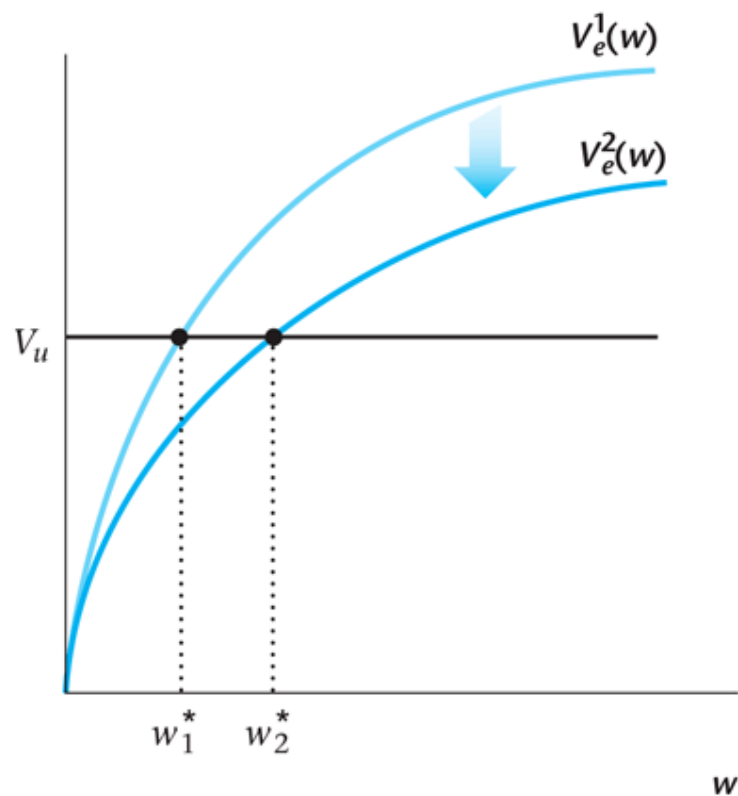
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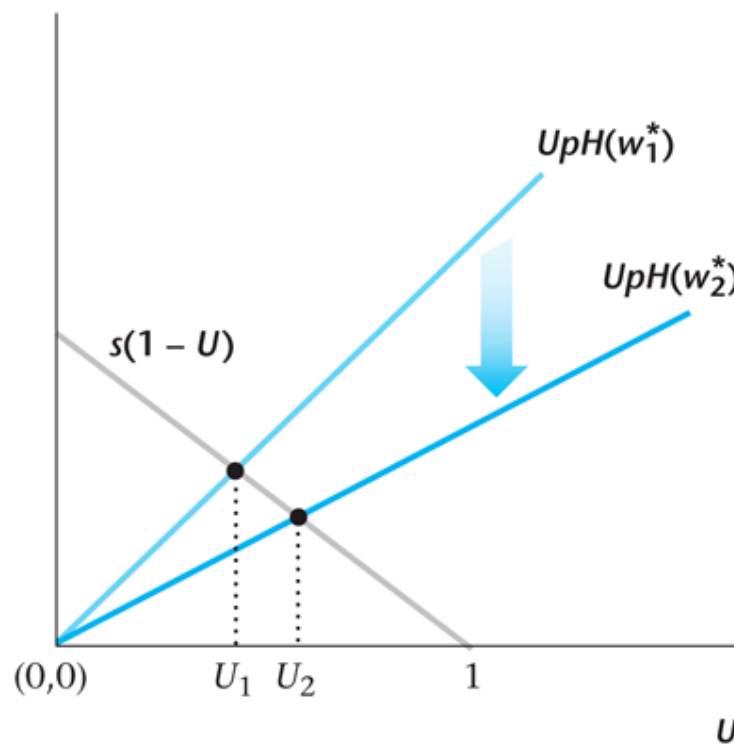
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Figure 17.15

An Increase in Taxes on Labor Income



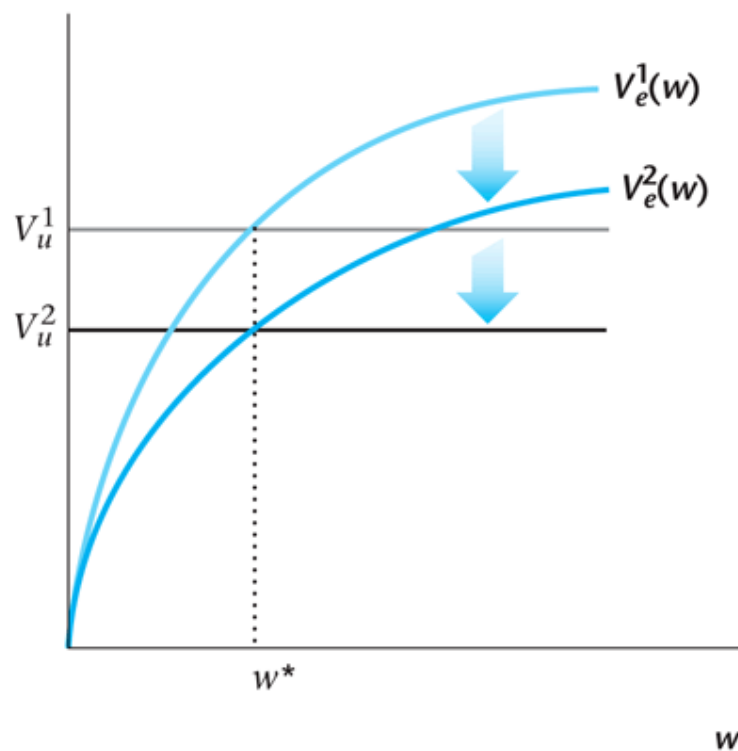
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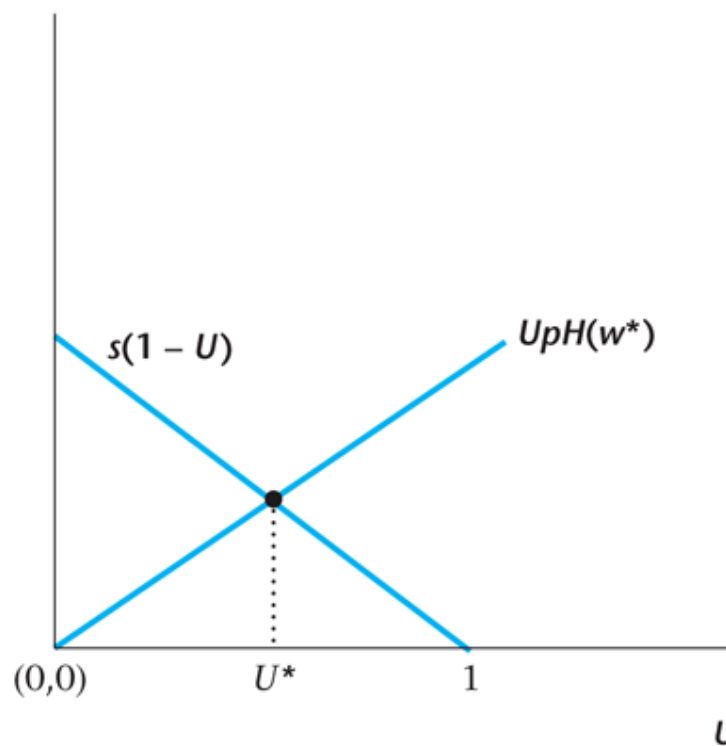
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Figure 17.16

Taxes on Labor Income and Unemployment



(a)



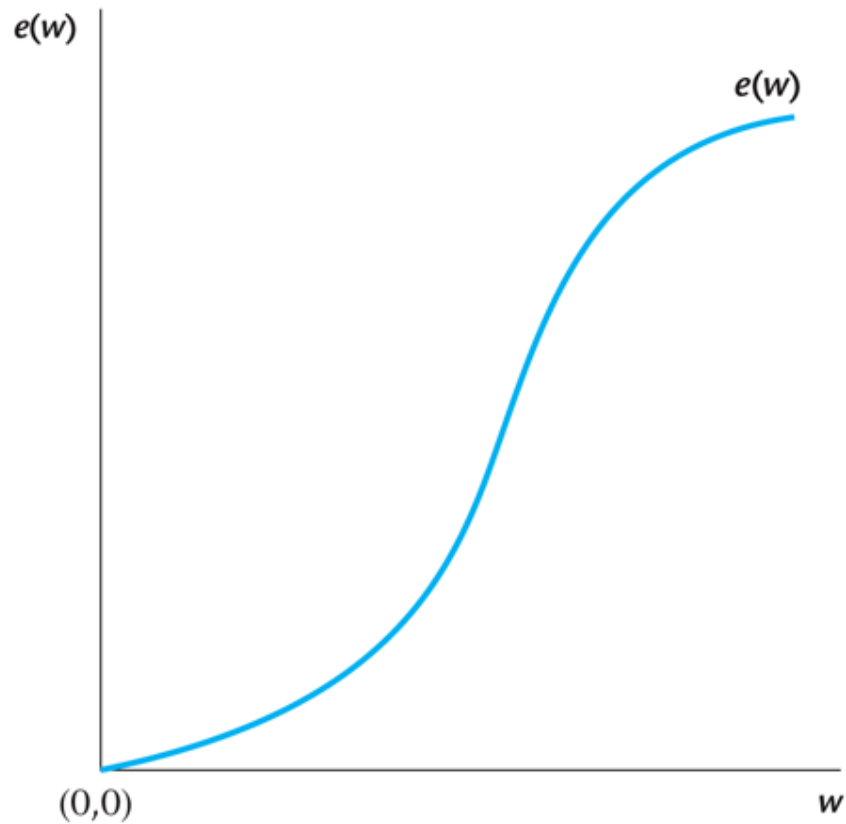
(b)

The Efficiency Wage Model

- Worker effort tends to increase with the real wage rate that they worker receives.
- This occurs because of *adverse selection* and *moral hazard*.
- This can produce a kind of stickiness in the real wage.

Figure 17.17

Effort of the Worker as a Function of His or Her Wage



Optimization by the Firm

When the firm optimizes,

$$e(w)MP_{e(w)N} = w.$$

Figure 17.18

The Demand for Labor in the Efficiency Wage Model

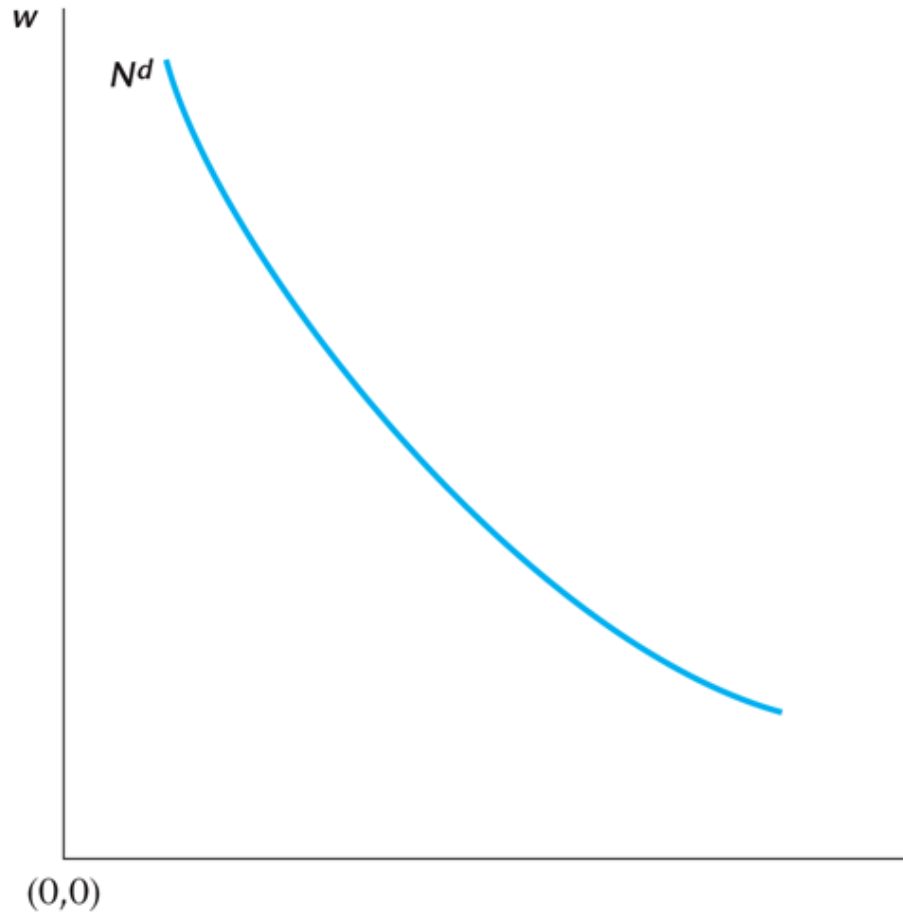


Figure 17.19

The Ratio of Effort to the Real Wage

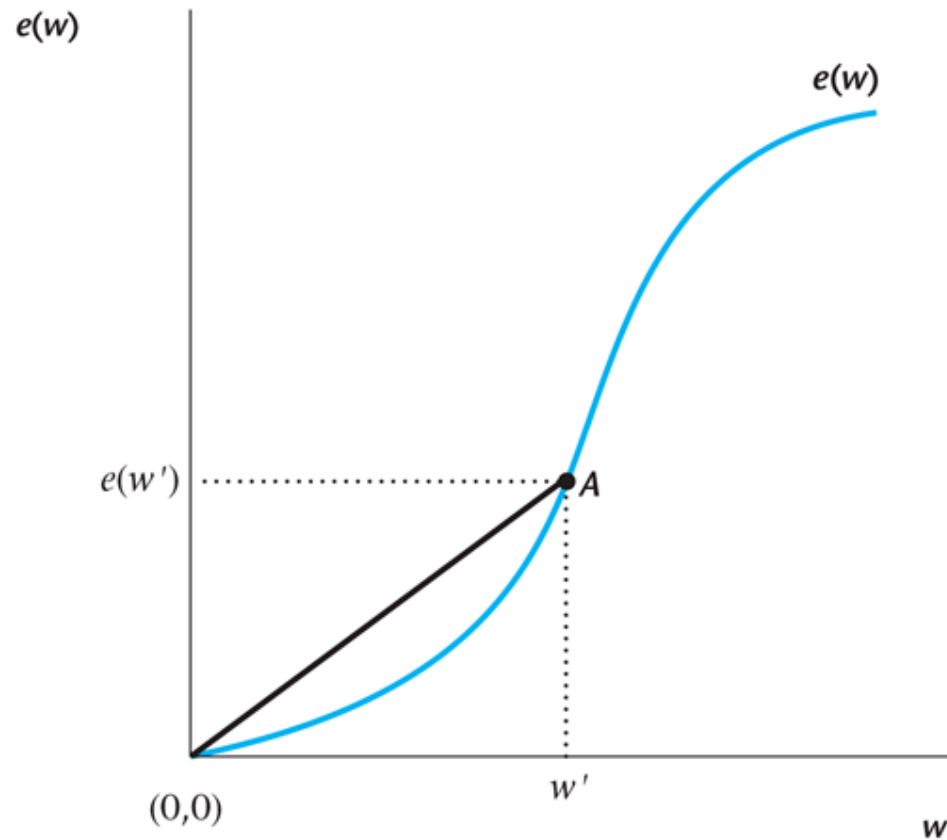


Figure 17.20

Determination of the Efficiency Wage

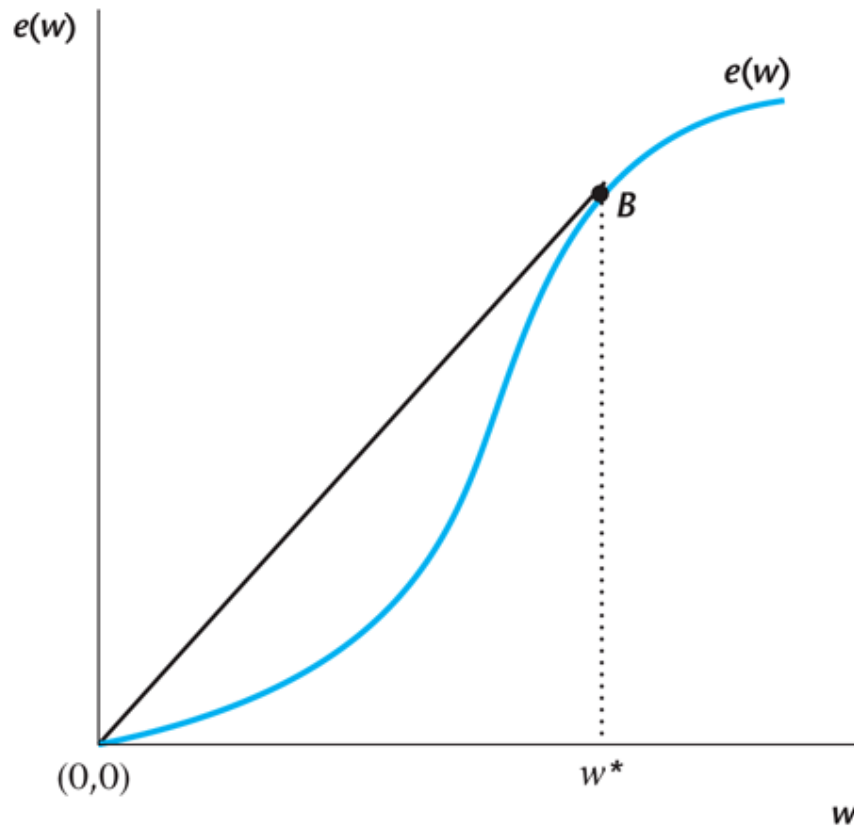
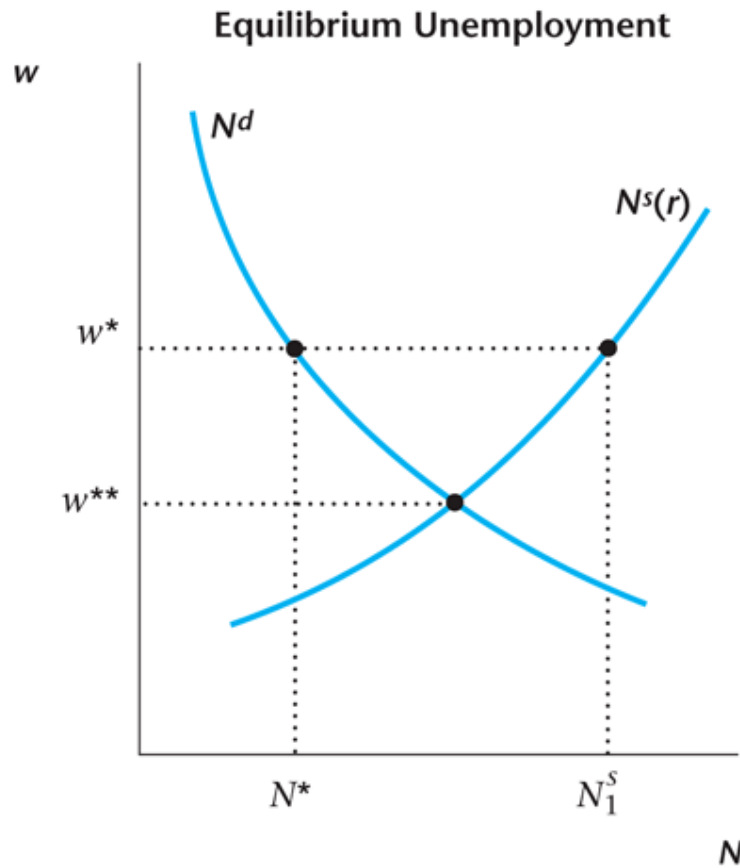
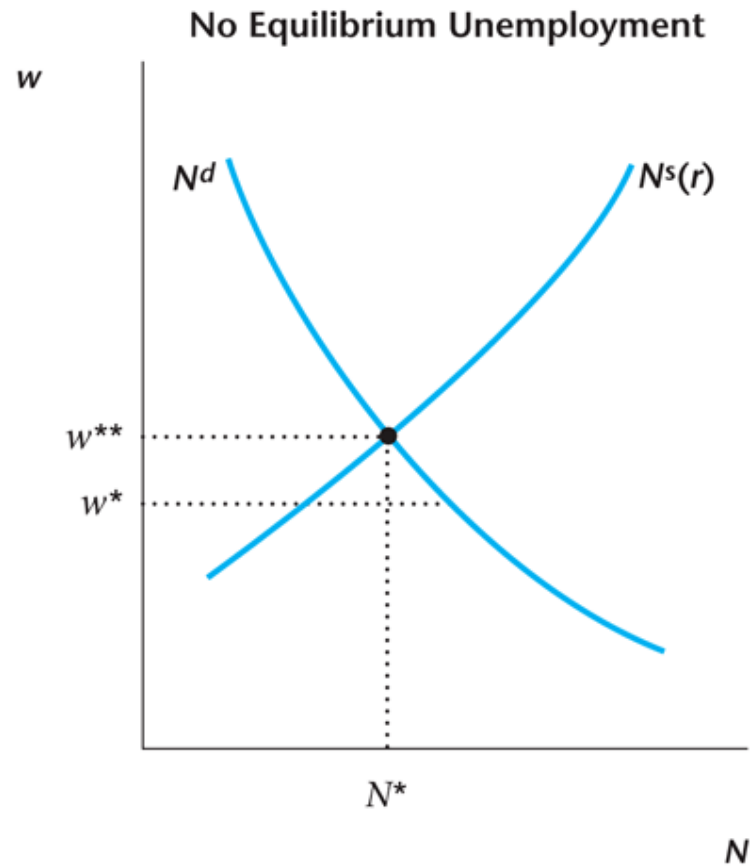


Figure 17.21

Unemployment in the Efficiency Wage Model



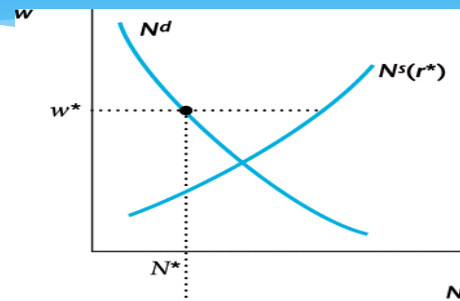
(a)



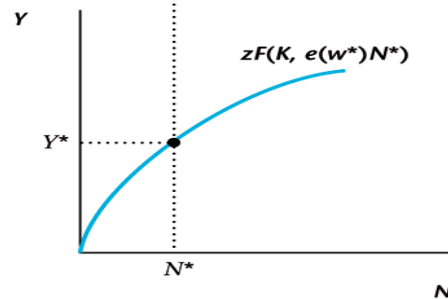
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Figure 17.22

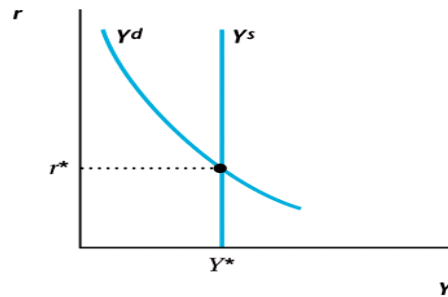
The Output Supply Curve in the Efficiency Wage Model



(a)



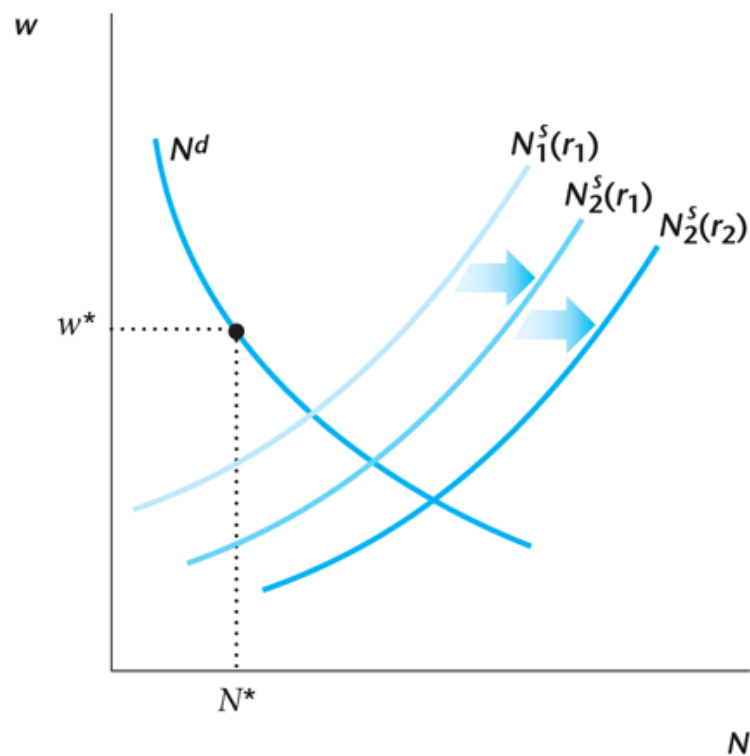
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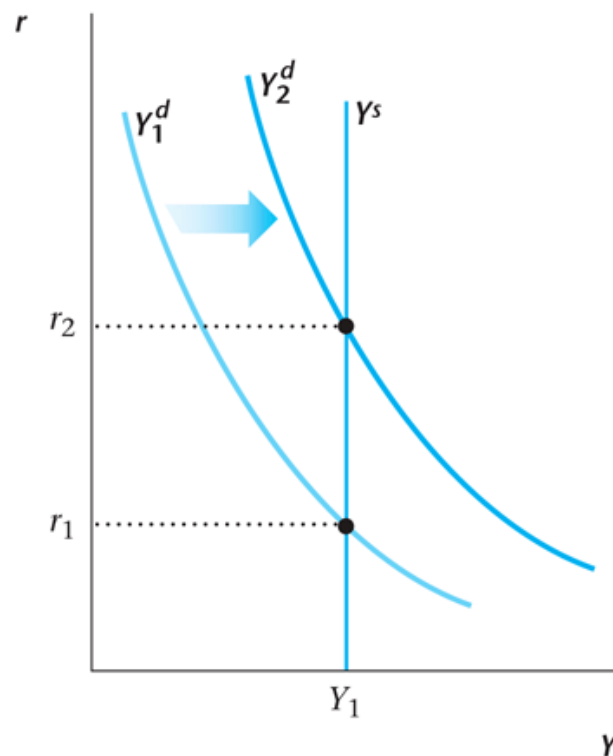
(c)

Figure 17.23

An Increase in G in the Efficiency Wage Model



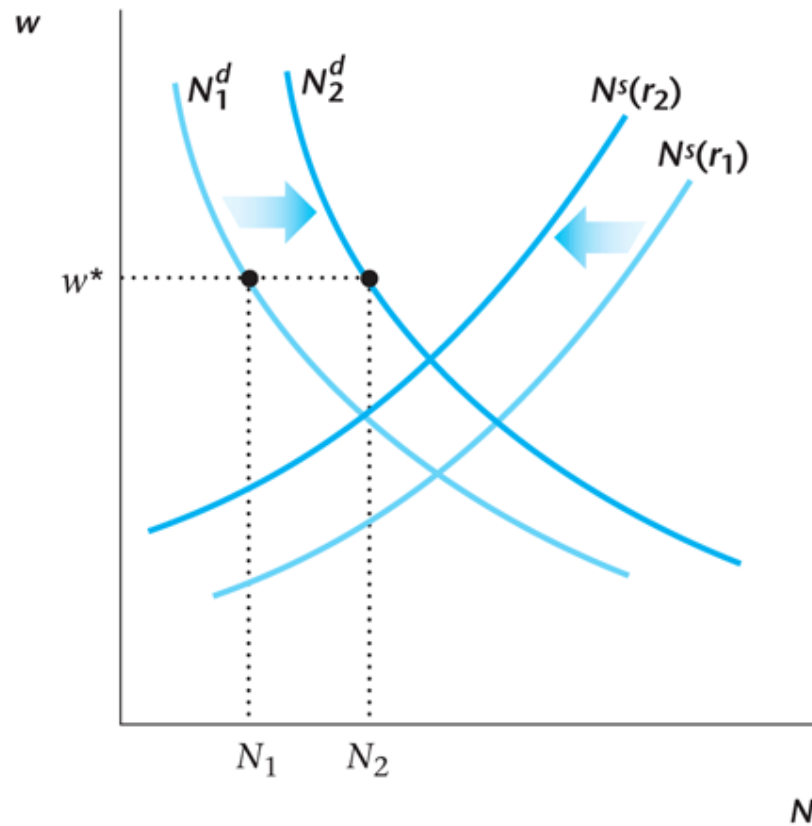
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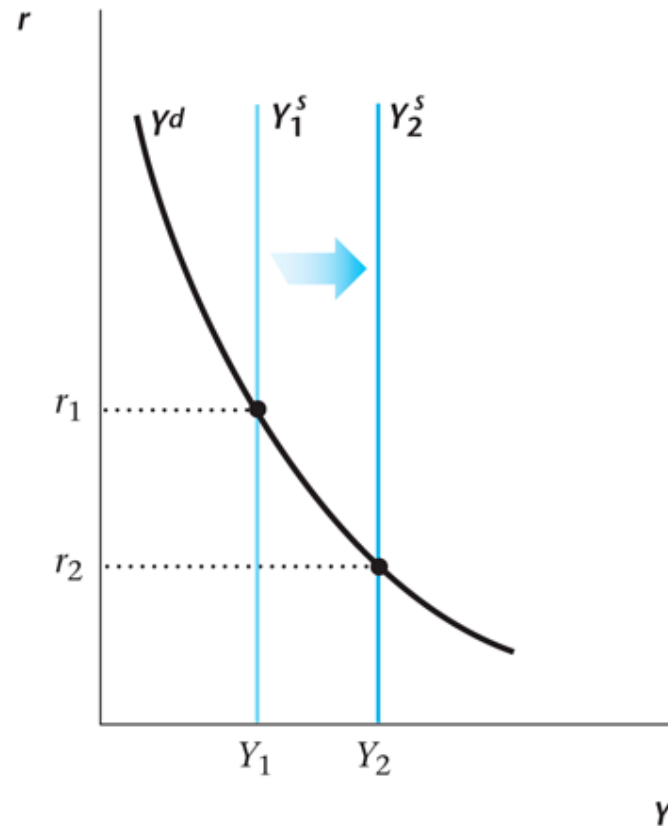
(b)

Figure 17.24

An Increase in Total Factor Productivity in the Efficiency Wage Model



(a)



(b)

Figure 17.25

An Increase in the Effectiveness of Monitoring by the Firm

